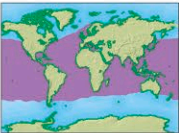


Xiphias gladius

Broadbill swordfish



Length Up to 15 ft (4.5 m)
Weight Up to 1,300 lb (590 kg)
Sex Male/Female
Status Least concern

Location Tropical, subtropical, and temperate waters worldwide



A popular food and game fish, this fast-swimming predator gets its name from its long, flat, swordlike snout (bill). These fish are extremely difficult to study in the wild, but fish in the stomachs of dead swordfish have obviously been slashed and the victims or bits of them swallowed whole as adults have no teeth. The broadbill swordfish is one of a handful of ray-finned fish that can maintain their brain and eyes at temperatures up to 59° F (15° C) higher than the surrounding water. This means they can stay alert and hunt efficiently even in cold, deep water. Swordfish (or perhaps other

billfish) have been known to slash into and even sink small boats, probably accidentally. Its chief predators are sharks, especially the mako and blue, and it is sometimes attacked by parasitic lampreys. Swordfish spawn in warm waters throughout most of the year, the months depending on the area.

SOLITARY HUNTER

Adult swordfishes usually hunt alone. Like tuna and marlin, they have similar proportions of the 2 distinct kinds of swimming muscles: dark red muscles are used for steady cruising, while faster-acting white muscles are used for short bursts of speed.



Sphyræna barracuda

Great barracuda



Length Up to 6½ ft (2 m)
Weight Up to 110 lb (50 kg)
Sex Male/Female
Status Least concern

Location Tropical and subtropical waters worldwide



The great barracuda is a formidable, fast-moving predator, with few natural enemies apart from large sharks. Its large head and powerful jaws conceal knifelike teeth. This fish remains relatively motionless, then lunges forward quickly to grab its prey. It has been known to attack swimmers, often causing serious injury. However, these mishaps have usually occurred in murky waters where a human limb or shiny jewelry can be mistaken for other fishes. Attacks have also been reported in clear waters when the fish has been provoked with a spear. It is curious and will swim up close to divers. The great barracuda is a popular game fish and is sometimes eaten, but large individuals can cause ciguatera poisoning—a

SOCIAL JUVENILES

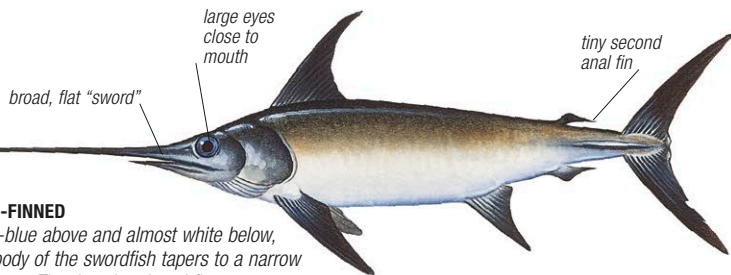
Adult great barracudas are, by and large, solitary cruisers, found typically at or near the surface of open tropical and subtropical seas, in warm and shallow waters. They have been caught occasionally at depths of 330 ft (100 m). The young, however, sometimes swim in small groups inshore, in shallow, sheltered reefs and mangrove swamps. This allows them to hunt in groups and also provides them with some protection from predators.



potentially fatal condition caused by toxins that barracudas ingest with their food.

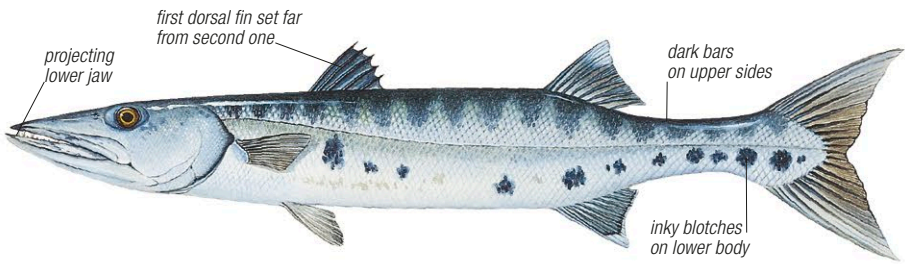
LONG BODY

The fish has an elongated body with a long snout. Its body length and associated musculature are specially adapted to produce short, rapid bursts of speed.



HIGH-FINNED

Gray-blue above and almost white below, the body of the swordfish tapers to a narrow tail base. The dorsal and anal fins are narrow and high, enabling fast turning.



Makaira nigricans

Blue marlin



Length Up to 14 ft (4.3 m)
Weight Over 1,985 lb (900 kg)
Sex Male/Female
Status Vulnerable

Location Atlantic



The blue marlin is renowned among anglers for the way it fights when caught. Hooked blue marlin breach the water surface with extraordinary leaps. The largest member of the billfish family, some individuals can weigh over 1,985 lb (900 kg). The upper jaw forms a large bill, which is apparently used to stun or kill prey. The lateral keels at the base of its tail and the pelvic fins set into grooves make this fish a powerful and fast swimmer, capable of explosive bursts of speed and long-distance travel. Although blue marlin may stray to greater depths, they prefer the warmth of surface waters, where they feed on other fishes such as mackerel, tuna, dolphinfish, and squid.

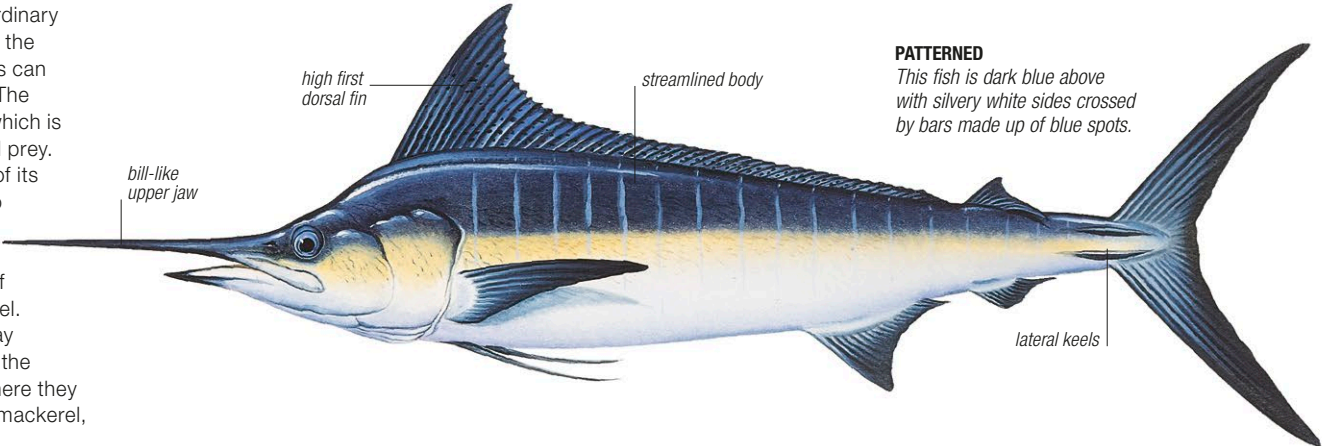


EGGS AND LARVAE

A single spawning produces millions of eggs, each ½ in (1 mm) in diameter. These possibly hatch within a week. Eggs and larvae are planktonic.

CONSERVATION

Game fishing can locally affect the numbers of this fish, but there is a much more serious threat in commercial fishing. A number of developing countries, whose offshore waters abound with this species, exploit the blue marlin as a popular source of food. Due to their size, adult marlin are generally safe from predators other than humans, but juveniles fall prey to other sea fishes such as tuna, mackerel, and sharks.



PATTERNED

This fish is dark blue above with silvery white sides crossed by bars made up of blue spots.